SOUND REVISION SHEET

Name _____

I. Match the words in column A with column B

Column A	**Column B**
Compression	Number of vibrations or waves per second
Decibels	Unit used to measure the frequency of a wave
Rarefaction	Type of wave formed by compressions and rarefactions
Hertz	An area of high pressure where particles are close together
Longitudinal	An area of low pressure, in which particles are far apart
Frequency	A measure of the energy of a wave
Amplitude	Unit used to measure the loudness of sound

II. Use the following words to fill in the blank spaces in the sentences below.

amplitude	decibels	frequency	solids	compressions	pitch	vibrations	more
loud	rarefactions	less	soft	medium	amount of energy	temperature	liquids

1. Sound is transmitted by _____ in a medium, such as air or water.
2. Sound waves are a longitudinal wave made up of a series of _____ and _____.
3. The _____ of the sound wave is related to the pitch of the sound.
4. A sound wave with high frequency has a higher _____ than a sound wave with lower frequency.
5. The _____ of a sound wave tells you how loud the sound is.
6. The loudness of a sound wave depends on the _____.
7. A loud sound has _____ energy than a soft sound and a soft sound has _____ energy than a loud sound.
8. Loudness is measured in _____.
9. A high decibel sound is a _____ sound. A low decibel sound is a _____sound.
10. The speed of sound depends on the _____ through which the sound wave is traveling and the _____.
11. Sound travels fastest in _____ and slowest in _____.

III. Multiple Choice

1. Sound waves travels fastest through which of the following? _____
   a) solids b) liquids c) gas

2. A sound wave's amplitude determines_____
   a) how high the pitch is b) how loud the sound is

3. A sound wave's frequency determines _____
   a) how high the pitch is b) how loud the sound is

4. If a sound is high-pitched and soft, the sound wave has _____
   a) high frequency and high amplitude c) low frequency and high amplitude
   b) high frequency and low amplitude d) low frequency and low amplitude

5. If a sound wave is high-pitched and loud, the sound wave has _____
   a) high frequency and high amplitude
   b) high frequency and low amplitude
   c) low frequency and high amplitude
   d) low frequency and low amplitude

6. If a sound is low pitched and soft, the sound wave has _____
   a) high frequency and high amplitude
   b) high frequency and low amplitude
   c) low frequency and high amplitude
   d) low frequency and low amplitude

7. If a sound is low-pitched and loud, the sound wave has _____
   a) high frequency and high amplitude
   b) high frequency and low amplitude
   c) low frequency and high amplitude
   d) low frequency and low amplitude

8. When someone tells you to turn down your ipod, they are is asking you to _____
   a) lower the frequency of the sound wave
   b) increase the frequency of the sound wave
   c) increase the amplitude of the sound wave
   d) lower the amplitude of the sound wave

IV. <u>Answer the following questions:</u>

1. Explain why sounds travel faster in solid and liquids than they do in gases.

2. The speed of sound in air is 330m/s. How long would it take for a sound to travel 5km?

3. List three ways to protect your hearing in a noisy workplace.

4. Explain the function of the following parts of the ear.
   a. Pinna

   b. Eardrum

   c. The ossicles

   d. Oval window

   e. Cochlea

   f. Auditory nerve

5. For one group of musical instruments, explain how sound is produced by them and list three examples of instruments in that group.